Durable Taste Schemas and Patterns of Cultural Choice

Abstract

Whether cultural schemas are malleable or durable is a key, but still unresolved, theoretical debate in cultural analysis. In this paper, I contribute to a recent line of work seeking to move this issue to the empirical terrain. To do so, I extract taste schemas from survey data on a representative sample of Americans collected in 2012 containing items on likes and dislikes of musical genres similar to those included in the 1993 General Social Survey culture module. Consistent with the durability position, I find remarkably similar taste schemas in 2012 as those uncovered in the 1993 General Social Survey data in previous work using comparable techniques. Going beyond previous work, I validate these schemas using independent behavioral evidence on musical listening choices and propose a novel way to think about the link between taste schemas and culture consumption behavior that can be generalized to other domains

Key words: Omnivorousness, Taste, Schemas, Cultural Choices, Univores

1. Introduction

1.1. The Durability versus Malleability Debate in Cultural Analysis

Are cultural schemas durable or malleable? This is a question on which cultural analysts drawing from different theoretical traditions split into different camps (Vaisey and Lizardo, 2016). One the one hand, we have analysts who talk about schemas as "deeply internalized" and "durable" cultural elements, and thus as having the potential to show continuity in relation to changes in the external environment (Bourdieu, 1990; Strauss and Quinn, 1997; Patterson, 2014). On the other hand, we have analysts who conceptualize schemas as highly responsive to changes in the external institutional environment and processes, and thus as "shifting," "multiform" (DiMaggio, 1997; Swidler, 2005; Sewell, 2004).

Pitched as a theoretical discussion, this debate runs the danger of becoming a purely scholastic issue. Here I break with this approach and instead follow the call recently laid out by Vaisey and Lizardo (2016) to move the issue into a more empirical terrain. I use the case of cultural taste schemas to examine the issue of whether we can observe continuity or transformation at the population level over a twenty-year period. The results reveal somewhat remarkable levels of schematic continuity in this domain, a pattern consistent with the relative durability thesis (Vaisey and Lizardo, 2016). However, partially consistent with the malleability position, the differences that we do observe can be readily traced to institutional transformations on the consumer and production side of musical genres, such as the aging of the Rock and Roll generation and the emergence of hip-hop and rap as a prototypical "pop" or "industry" genre.

1.2. The Case of Cultural Taste Schemas

Cultural taste schemas, especially those linked to the underlying association shared by people with respect to musical genre categories, represent a strategic research site with which to address the issue of schematic durability. This is for two reasons. First, both theoretical and methodological developments with regard to extraction of latent schemas from observed responses in survey data have used this case, and the associated data source (the 1993 General Social Survey), as their primary testbed (Goldberg, 2011; Boutyline, 2017). This means that we have a set of relatively well-established results documenting population-level variability on taste schemas in the American public at this time characterizing the nature of each evaluative schema, thus facilitating comparison.

Second, the domain of musical taste is different from other institutional settings such as politics. In the political domain, we might expect continuity and durability in people's conceptions to be fixed by the relative durability of external institutional arrangements (e.g. the two-party system in the U.S.). And in fact, this is precisely what we observe. In a landmark study, Baldassarri and Goldberg (2014) find remarkable continuity in the ways Americans organize their understanding of political issues at the population level across twenty years (although they could not engage in apples to apples comparisons of the same items as we do here). A durability theorist might see this as strong evidence for their position, but a malleability theorist could point to continuity (even with change at the margins) in external institutional arrangements as a competing explanation. In the case of politics, therefore, continuity seems over-determined by external and internal mechanisms, making it hard to adjudicate between the durability and malleability positions.

Musical taste, by way of contrast, represents a case in which radical transformations have occurred in terms of production, distribution, modes of consumption, as well as definitions of genre popularity and "aesthetic mobility" (Baumann, 2001) since the early 1990s. Thus, if we were to expect, consistent with the malleability thesis, that schemas respond to radical structural and external changes in a given domain, then the domain of musical evaluation would be an ideal case in which to observe such transformations. On the other hand, observing relative continuity in this domain would support a strong version of the durability thesis (Vaisey and Lizardo, 2016). In this paper, I set out to do just that by analyzing twenty distinct musical genres—ranging from Classical and Opera to Rap and Heavy Metal—to see how they cluster together in people's minds. Specifically, I anticipate that core anchors like Rock, Pop, and Rap will fundamentally organize taste schemas, while others like Heavy Metal will remain polarized.

1.3. Terminology and Approach

It is important to get clear on the use of certain specialized terms that will recur through the manuscript. In what follows, I refer to the latent schema presumed to be responsible for a given pattern of observed responses to evaluative questions in a cultural taste survey as a *taste schema*. Persons who share the same (or similar) latent taste schemas are members of a *schematic class*. Following (Boutyline, 2017), I refer to *schematic class analysis*, as a set of procedures designed to classify respondents based based on their sharing latent schemas governing their observed (usually evaluative) ordered responses to a given set of objects in survey data (such as likes and dislikes of musical genres in the case of cultural taste).

Traditional methods such as factor analysis, latent class analysis, or principal component analysis attempt to group respondents into classes based on differences in their observed responses. The assumption is that there is a one-to-one correspondences between latent constructs driving the responses (such as an attitude) and observed response patterns. Persons at the same level (or in the same category) of the latent construct will produce similar observed responses. Schematic class analysis, by contrast, exploits the distinction between a latent schema, which is an unobserved pattern of associations as to what goes with what or what is the oppossite of what, and an evaluation which is a "pro" or "con" (or in mixed) attitude directed to an object (Goldberg and Stein, 2018). Two people can agree about the way they think two objects are associated (e.g. thinking that Classical music is completely opposed to Country music), while being at different ends of the evaluative scale with respect to the same pair of objects (one person loves Classical and hates Country, while the other is a Country fan who thinks Classical is boring).

The above implies that persons can provide different observed evaluative response patterns even when they share the same latent schema with regard to the relation between a set of objects. The goal of schematic class analysis is thus to partition respondents based on these latent schemas exploiting specifiable relations between observed response patterns. Boutyline (2017) argues that this can be done under the assumption that, when two different observed response patterns come from the same schema, the two response sets will be related via a finite set of transformations, such as inversion or rescaling (or both).

Correlational Class Analysis (CCA) is a variant of schematic class analysis that uses the absolute value of the Pearson correlation coefficient between the response vectors of different respondents as a schematic similarity criterion. The idea being that the observed response patterns of two people have a large absolute correlation, they are likely to have been produced by transformations of the same latent schema (Boutyline, 2017). Like relational class analysis, CCA partitions the resulting (weighted) network of respondents, where links are based on the absolute value of the correlation, using a modularity maximization algorithm. Boutyline (2017) shows that the correlation approach outperforms other ways of computing response pattern similarity, and thus eliciting schematic class membership, such as Goldberg's 2011 use of the "relationality" measure in relational class analysis. Given this, I use correlational class analysis as the main approach to extracting schematic classes in what follows.

1.4. Previous Work Using Schematic Class Analysis

Schematic class analysis, whether based on relationality or the correlation distance, has been used in a variety of applications to extract schematic classes from mass survey responses. These include political belief logics (Baldassarri and Goldberg, 2014; Daenekindt et al., 2017), patterns of thought about the markets and the economy (DiMaggio and Goldberg, 2018; Metz-McDonnell et al., 2018), and taste schemas about the link between science and religion (DiMaggio et al., 2017). In addition, recent work by Daenekindt (2017) analyzes the structure of aesthetic dispositions among a sample of museum-goers, a topic close to that dealt with here. In this respect, schematic class analysis has already made major contributions to the social science literature across a wide range of topics.

1.5. Contributions of the Present Work

Previous work using schematic class analysis suffers from validation limitations. In particular, previous studies are forced to rely on inspecting differences in the correlations between items within classes (internal validation) or the correlation between schematic class membership and demographic characteristics to both validate schematic class assignment, and make inferences about the possible effect of schemas on behavior. Goldberg (2011) used internal, and demographic validation strategy in the original study using 1993 General Social Survey data.Baldassarri and Goldberg (2014) used a combination of internal, demographic, and criterion-based strategies (looking at how predictors differ in predicting partisanship across schematic classes). In a similar way DiMaggio and Goldberg (2018) and DiMaggio et al. (2017) use a combination of internal, demographic, and substantive validation strategies.[1]

This study adds to and extends previous work by using both the standard internal validation approach (based on within-class inter-item correlations) and also behavioral (or criterion) validation using a measure of choice behavior directly related to the domain in question, namely, music listening habits. In this respect, my strategy comes closest to the that pursued by Baldassarri and Goldberg (2014). This allows me to determine empirically the behavioral potential of schematic classes in a way that establishes external (predictive) validity within the musical taste domain. I incorporate insights from recent work in the analysis of item correlation networks (Epskamp et al., 2012), and link recently developed formal measures of item position in the taste schema with behavioral predictive criteria. Promising results obtained with this approach recommend it as a useful strategy for future schematic class analysis work.

2. Data and Variables

In the summer of 2012, I fielded a survey that was (partially) designed to replicate the General Social Survey 1993 Culture Module. This last is now a canonical dataset, providing the empirical basis for large and variegated set of analyses (and re-analyses) in the sociology of taste (Boutyline,

[1]Note that this discussion is not related to validation of the *number* of classes extracted by schematic class analysis methods such as RCA or CCA. In this respect, Boutyline (2017) have used multiple group SEM, and DiMaggio et al. (2017) and DiMaggio and Goldberg (2018) have used network randomization methods coupled with cluster validation (e.g. gap statistic) techniques.

2017; Bryson, 1996, 1997; Goldberg, 2011; Han, 2003; Okada, 2017; Tampubolon, 2008, among others). It can be said without exaggeration that the bulk of our knowledge of the cultural taste patterns of the American population, both in terms of schematic form and actual content, are due to this data source. As such, the addition of a more contemporary source on the musical taste patterns of the American population constitutes a valuable addition on its own right (Lizardo and Skiles, 2016a,b, 2015).

The data used in this paper were collected by Survey Sample International (SSI), a private firm that specializes in sampling, data collection, and analysis. SSI managed recruitment and participation invitation tasks to generate a panel of adults from which our working sample was drawn. Survey respondents were selected from the panel for participation on the basis of age, gender, race, education, and geographic region to approximate a sample representative of the U.S. population, $N = 2,250$.

Like General Social Survey 1993, SSI 2012 included items assessing respondents' likes and dislikes (as well as a middle category of "mixed feelings") for 20 categories of musical style: (1) classical or symphony and chamber, (2) opera or operetta, (3) jazz, (4) Broadway musicals or show tunes, (5) mood or easy listening, (6) big band or swing, (7) classic rock or oldies, (8) country, (9) bluegrass, (10) folk, (11) hymns or gospel, (12) Latin or Spanish/salsa, (13) rap or hip hop, (14) blues or R and B, (15) reggae, (16) top 40 or pop music, (17) contemporary rock, (18) indie or alternative rock, (19) dance or club/electronic, and (20) hard rock or heavy metal. These responses are the basis of the schematic class analysis.

The survey also asked respondents to report in a separate item whether they regularly *listen* to each one of the genres listed. Responses to these items allow us to examine behavioral expression (and variation) across schematic classes separately from the items used to elicit schematic class membership.

3. Correlational Class Analysis

I follow the same data analysis procedure as Boutyline (2017), with two modifications. The first is relatively minor; given the nature of the taste evaluation items as limited ordinal response items, I use the polychoric correlation coefficient as a "drop-in replacement" (Boutyline, 2017, 386) for the Pearson correlation.[2]

The second is less trivial and has to do with the way I treated the "not familiar with this genre" responses to genre evaluation questions. Goldberg (2011) recoded these to the "mixed feelings" (middle) response for all respondents who offered this response for six genres or less (dropping those who were not familiar with seven or more genres). Boutyline (2017) followed this same coding strategy for purposes of strict replication.

Since "not familiar" responses are a type of missing data, I follow recently recommended best practice and use multiple imputation rather than fixed value recoding. This strategy is more efficient and does not require us to choose an arbitrary cutoff to decided how many "not familar" responses are sufficient to drop a respondent from the analysis. The variables used in the multiple imputation procedure consisted of the other genres for which the respondent did provide a valid taste response. To avoid issues with median aggregation across multiple imputations, I utilized a single robust imputation via predictive mean matching. This was performed using the `mice`

[2]Polychoric correlations were computed using the function `cor_auto` of version 1.5 of the `R` package `qgraph` (Epskamp et al., 2012).

package in R, generating a complete dataset for the correlational class analysis without creating artificial fractional ordinal values. The distribution of "not familiar" responses is highly skewed: $1,511(67\%)$ of respondents claimed to be familiar with all genres; another 11% reported only one "not familiar" response. By the time we get to six "not familiar" responses we have reached 95% of respondents. Thus, only 5% of the analytic sample contains seven or more imputed responses. After dropping respondents who did not provide taste responses on any of the 20 genres ($N = 21$) and dropping respondents who provided the same response across all genres ($N = 72$), which correlational class analysis would treat as its own "degenerate" schematic class, the final analytic sample is $N = 2,183$.

3.1. Correlation Networks for Schema Extraction

As is now standard, I present the resulting schematic classes as correlation networks (Boutyline, 2017; Goldberg, 2011; Epskamp et al., 2012), with the edge weight set to the estimated correlation for each genre pair for members of that class and the node color set to the assignment of each genre to one of four latent factors based on the binary consumption items (results from which are shown in figure 1). Correlation networks for each taste schema are shown in Figures 2a- 2e.

3.1.1. Expected Influence Score

To quantify the importance of the evaluation of different genres within each class, I follow the approach recommended by Robinaugh et al. (2016) for analyzing node centrality in signed correlation networks and rank each genre in each taste schema by its *expected influence*. Fundamentally, Expected Influence is a form of weighted degree centrality (Barrat et al., 2004), given by:

$$EI_{gs} = \sum_{j=1}^{N} r_{gjs} \tag{1}$$

Where EI_{gs} is the expected influence of the g^{th} genre in the s^{th} taste schema and r_{gjs} is the value of the polychoric correlation coefficient between the taste evaluations for genre g and genre j among respondents who share taste schema s. Following Robinaugh et al. (2016), we can interpret this metric through the lens of a "spreading activation" model. In this framework, an increase in the evaluation of a focal node probabilistically causes an increase in positively tied nodes and a decrease in negatively tied nodes. Therefore, genres with the highest EI (those with strong, positive ties to many other genres) are expected to exert the most structural "influence" over the state of the overall network. To facilitate direct comparison across schemas, centrality measures are standardized to Z-scores. Figure 2f shows the rank of each genre according to EI_g for each taste schema.[3]

4. Results

4.1. Number of Taste Schemas and Class Sizes

Subjecting the data to CCA, yields a solution (setting weights in the respondent by respondent correlation network below $p < 0.01$ to zero) that partitions the sample into five schematic classes.[4]

[3] EI_g is computed using the `expectedINF` function included the R package `networktools` (Jones, 2018).

[4] To perform correlational class analysis, I used the routines included in the R package `corclass` (Boutyline, 2016).

This is similar to the results reported in Boutyline (2017) applying the same method to General Social Survey data, who found four schematic classes characterizing the American population in 1993. The classes that I obtain using the SSI 2012 data are somewhat uneven in size, with $N = 481(22\%), 369(17\%), 574(26\%), 419(19\%)$, and 340 (16%) respectively.

4.2. Taste Schemas in 2012 are Similar to Those Uncovered in 1993 General Social Survey

The most striking result is that the schematic classes evident in the 2012 data are very similar to those uncovered by Goldberg (2011) and Boutyline (2017). This suggests a rather remarkable level of schematic continuity in the American population (at least when it comes to musical genre evaluation), even in the face of large scale transformations in this domain. This supports the main implication of the durability position in cultural analysis (Vaisey and Lizardo, 2016).

First, one of the schematic classes (Figure 2a), which is similar to Boutyline's "anything (but) country," is characterized by an opposition of country music to a large clique of positively connected genres (with country featuring a long tie to Gospel/Religious). The key difference between my results and those reported by Boutyline (2017) is that country does not have a lot of negative connections (as in General Social Survey 1993) featuring instead mostly absent or very weak connections. It is clear, however, that members of this schematic class decouple (if not oppose) their evaluations of Country from that of other genres. In addition, just like Boutyline (2017, Figure 10D), I find that Gospel/Religious music is also relatively decoupled from the evaluations of the main cluster in this taste schema, featuring a number of weak connections. The main difference is the status of Hip Hop/Rap as a peripheral genre in this schema, which was not the case in the 1993 data. In fact, as the first panel of Figure 2f shows, in these data this schema would be best referred to as "Anything (but) Country or Rap" as these are the least influential genres in the schema. In terms of network centrality, 'highbrow' genres (Classical, Jazz) and standard rock act as the core structural anchors of this schema, while Country and Rap are highly peripheral. Overall, however, we see a strong continuity pattern in relation to Boutyline's results.

Second, another of the schematic classes bears a strong resemblance to that which Boutyline (2017, Figure 10) referred to as "Anything (but) Heavy Metal" (Figure 2b). My results, however, bring out something that was already evident in Boutyline's results: The fact that this schema contrasts the evaluative judgment of both Metal *and* Hip Hop/Rap (whose evaluations are strongly connected) against (mostly) everything else. However, as shown in Figure 2f, it is fair to refer to this as a schematic class opposing heavy metal univores against eclectics who prefer other genres, as Metal has the least influence in the correlation network (followed by Rap). Structurally, its core network anchors are Classic Rock, Rap, and Bluegrass. Interestingly, Rap has high centrality here, meaning that whether respondents tolerate or reject Rap is a massive polarizing divider that dictates the rest of their network. Overall, we once again observe a strong continuity pattern in relation to Boutyline's results.

Third, just like Boutyline (2017) and Goldberg (2011) I uncover a taste schema that contrasts "traditional" (Country, Gospel/Religious, Bluegrass, Folk, Classical, Big Band, Opera) against "contemporary" genres (Figure 2c). Like Boutyline and Goldberg, I find that Contemporary Rock, Metal, and Rap and Reggae (in addition to Pop, Dance, and Alternative, which were not included in General Social Survey 1993) are squarely on the "contemporary" side of the divide. I also find that Latin and Country are in an intermediary ("brokerage") position between tradition and contemporaneity, possibly pointing to heterogeneity within these genres keyed to this distinction (e.g. the "folk" versus "pop" look in Country (Peterson, 2013, 52)). The main difference concerns the position of Jazz, which in Boutyline's analysis fell on the side of contemporaneity. In 2012, this genre is more unambiguously on the tradition side, suggesting classificatory consequences from Jazz's move to a traditionalist "conservatory" genre (Lena, 2012). Organized heavily around mainstream pop and contemporary music, Pop is the absolute central load-bearing pillar of this schema, while Opera acts as a highly central negative anchor. Overall, it is clear that the "Contempo/Trad" schema is a durable feature of the American taste public.

Fourth, like both Boutyline and Goldberg, my analysis also reveals a schematic class composed of mostly strong and positive, as given by the thick gray edges, and a few null or very weak negative correlations. Like these predecessors, I label this taste schema "Omnivore-Univore" although this label is partially misleading, if it is taken to mean that this is the only schematic class capable of producing a version of omnivorousness at the behavioral level; a better assumption to make is that this is the class with the potential for the most extreme forms of omnivorousness in terms of actual consumption choices, a claim we will evaluate below. One thing that was already evident in Boutyline (2017, figure 10) and Goldberg (2011, Figure 3A) but that it is clear in these results, is that the peripheral status of Metal on this schema in relation to the rest of the genres. This is confirmed in Figure 2d, which suggests that the prototypical univore in this class is a "Metalhead" opposed to persons displaying a favorable attitude to a wide range of musical styles. Structurally, the network revolves around accessible, easy-to-digest music (Easy Listening, Pop) and roots music (Bluegrass), with harder genres like Metal pushed completely to the periphery. As before, what is striking here is the similarity of these results to those obtained by Boutyline and Goldberg using data collected about a quarter of a century ago.

Finally, unlike Boutyline, but like Goldberg, I uncover a schematic class bearing a strong resemblance to the one Goldberg labeled "High-Low" in General Social Survey 1993 using relationality as a distance criterion, but which Boutyline failed to uncover using the correlation distance in the same data. However, unlike Goldberg, who uncovered a largely binary partition, members of this class seem to divide genres according to a three-part distinction between genres, strongly linking traditional genres popular in rural areas (e.g. Country, Folk, and Bluegrass), separating these from traditional urban "easy listening" fare popular with older respondents, and both of these from contemporary genres. These results point to a taste schema shared by persons that engage in a more traditional style of classification across meta-genre discourses similar to the traditionalists discussed in Van Eijck (2001). Accordingly, I label this schema "Contempo/Folksy/Easy."

4.3. Behavioral Validation of Taste Schemas Using Music Listening Choices

As noted earlier, previous work using schematic class analysis generally lacks a behavioral criterion with which to validate results. Analysts usually have to rely on indirect validation procedures, such as looking at the correlation between taste schemas and demographics (Goldberg, 2011; Baldassarri and Goldberg, 2014) and inferring subsequent behavior from what is known about the correlation between demographics and action. The 2012 survey asked respondents to not only evaluate genres but also to list whether they listened to those genres or not. This information provides a source of behavioral validation for the schematic classes.

4.4. The Omnivore-Univore Taste Schema Displays the Most Ommnivorousness

I begin by looking at the distribution of the number of consumption choices across taste schemas. This is shown in Figure 3a. If the omnivore/univore schema lives up to its label, then people who share this schema should be over-represented among those who listen to a large number of genres and should be relatively under-represented among people who do not listen to many genres. Figure 3 shows that this is indeed the case, as the blue line corresponding to the univariate distribution of genres chosen is flatter than that corresponding to the other schemas. However, the figure also shows that all taste schemas have some behavioral potential for both omnivorousness and univorousness. In this sense, people who share the same taste schema can end up on either side of the omnivore/univore consumption divide.

4.5. Centrality in a Taste Schema Predicts the Link Between Omnivorousness and Evaluation

More convincing evidence for the criterion validation of the schematic classes would be to link the position of each genre in the schematic network to cultural consumption as a behavioral criterion.

The validation strategy here adapts the design used by Robinaugh et al. (2016). In their work, Robinaugh et al. first used simulations of a "spreading activation" model to demonstrate that the Expected Influence (EI) of a node correctly predicts how much effect its change in activation would have on the rest of the network. They then empirically validated this by showing that a node's EI tracked its behavioral predictiveness over time. Adapting this logic to cross-sectional data, we can define a behavioral measure of importance for genre j, $importance_j$, as the correlation between a respondent's evaluation of genre j and their overall behavioral omnivorousness (the total count of genres they actually listen to). If the spreading activation model holds as a useful methodological heuristic, the structural centrality of a genre (EI_j) should track this behavioral measure of importance.

Now imagine, the same process but this time we "activate" (by producing a positive evaluation) a genre that is peripheral in the correlation network of that schema (like Country or Hip Hop in Figure 2a). Because such genres feature mostly null or negative links with other musical styles, then we should expect that the evaluation of those genres should be decoupled (or in the case of negative correlations lead to a devaluation) of the other genres in the schema. In this case, there is no spreading activation and no empirical expectation that evaluations of these genres should be linked to behavioral omnivorousness.

Because evaluation is usually strongly linked to consumption behavior, then liking lots of genres should result in an increased likelihood of actually listening to a lot of musical styles under normal conditions (displaying omnivorousness in consumption behavior), and liking only a few should make you a univore.

If this imagery of the way taste schemas link to behavior is on the right track, then we should expect that omnivorousness should be positively correlated with the positive evaluation of the *most central genres* in each taste schema. In the same way, the number of genres actually listened to should have a lower correlation with the evaluation of genres that are peripheral in the correlation network for that schema. This is a cognitively plausible, and empirically verifiable, way of conceptualizing the link between taste schemas and consumption behavior, and one that should obtain if schemas are a valid predictor of such behavior.

Results of an analysis meant to evaluate this hypothesis are shown in Figure 3b. In the figure the y-axis gives the Spearman rank correlation between an indicator of behavioral omnivorousess—number of genres that people who share that schema listen to—and the ordinal taste evaluation item for that genre (higher values coded as liking that genre), re-scaled to the variance for that taste schema. The x-axis is the importance of the genre in the corresponding taste correlation network (as given by the expected influence (EI_{gs}) score; see figure 2f).

As expected, we see a positive correlation between these two quantities within all five taste schemas, with the correlation being strongest ($r = 0.68$) for the omnivore/univore schema. The upper-right of each scatterplot shows those genres for which there is a stronger than expected correlation between choosing a lot of genres and actually liking that genre. This in effect, gives us the expected taste profile for the omnivores in each class, in effect providing a schematic basis for the claim, now fairly standard in the sociology of taste, that there are multiple "varieties" of omnivorousness at the level of consumption (Coulangeon, 2015; Ollivier, 2008; Savage and Gayo, 2011).

Thus, Anything (but Country or Rap) omnivores are expected to combine Classical, Metal, Jazz, Alternative and Contemporary Rock, Blues, and Latin; Anything (but Heavy Metal) omnivores should be more likely to combine Contemporary Rock, Classical, Blues, Dance, Pop, Folk, and Alternative; Contempo/Trad omnivores are expected to listen to all forms of Rock, Metal, Reggae, Pop, Dance, and Blues; Omnivore/Univore omnivores like Reggae, Rap, Jazz, Blues, Dance, Pop, Contemporary Rock, Musicals, and Big Band; finally, Contempo/Folksy/Easy omnivores are mostly attracted to a variety of Folksy and easy listening genres, such as Classic Rock, Classical Music, Opera, Broadway, Jazz, Latin, Bluegrass, and Blues. This constitutes suggestive evidence for the behavioral validity of the correlation networks extracted from each taste schema and the spreading activation interpretation of the link between the resulting schemas and patterns of cultural choice.

The lower-left quadrant of each scatterplot shows genres for which there is a lower than expected correlation between behavioral omnivorousness and the evaluations of genres that are peripheral to each taste schema. This in effect, gives us the expected consumption profile of *univores* in each schema, providing evidence for the oppositional characterization of the nature of each schemas extracted from the correlation networks. Thus, Anything (but Country or Rap) univores tend to consume either Country or Rap, Anything (but Heavy Metal) univores are attracted to Metal and Rap, Contempo/Trad univores are "restrictive highbrows" attracted to Classical, Opera, and Religious music, Omnivore/Univore univores will be most likely to consume Metal, and Contempo/Folksy/Easy univores will gravitate toward Metal, Rap, or Country.

5. Discussion and Conclusion

This paper began with the debate over the malleability versus the durability of schemas in cultural analysis. To shed empirical light on this issue, I revisited the relatively well-studied issue of heterogeneity in evaluative schemas for musical genres in the American population (Goldberg, 2011). Using recently collected survey data comparable to the General Social Survey 1993 culture module, I find that taste schemas of Americans in 2012, bear a strong resemblance from those uncovered in previous work using comparable techniques (most closely Boutyline (2017)). I interpret these results as providing strong evidence that, at least at the population level, there is strong continuity and durability in the way Americans organize their understandings of an important cultural domain. These findings deepen and extend recent arguments arguing against extreme versions of the schematic malleability thesis, showing that we find relatively strong levels of population-level durability in a variety of cultural patterns (Vaisey and Lizardo, 2016). At the same time, the minor structural shifts observed in 2012—such as the repositioning of Hip Hop/Rap as a highly central organizing anchor in certain schemas, and the migration of Jazz toward traditionalism—highlight that schemas are not entirely static. Instead, they exhibit bounded malleability, adapting to macro-level institutional transformations in music production and consumption without losing their core structural integrity.

This paper also contributes and extends previous work in survey-based schematic class analysis by attempting to validate the results of the resulting schema partitions using behavioral criteria linked to the domain in question (in this case, patterns of musical listening choices). Building on the work of Goldberg (2011), who utilized network metrics like the clustering coefficient to examine schema networks, I showed that we can leverage modern techniques developed for Exploratory Graph Analysis (using the Leiden algorithm) to go beyond eyeballing (Epskamp et al., 2012; Jones, 2018). In particular, I showed that a simple measure of vertex centrality referred to as "expected influence" (Robinaugh et al., 2016), does a good job of capturing the genres most central in each schema. The validation results reveal two sets of notable findings.

I found that the taste schema featuring the lowest number of negative links among evaluative items (labeled Omnivore-Univore in previous work) is indeed the most omnivorous in terms of their actual likelihood to listen to a wide variety of musical styles. This supports the idea that at least in some cases, behavioral predictions can be made from "eyeballing" the correlational network corresponding to a given schema. However, beyond these broad predictions, it is perilous to conflate the evaluative structure of a schema with its behavioral potential. For instance, I find that all taste schemas have potential for a form of omnivorousness, but this cannot be ascertained from a global assessment of the correlation structure. Instead, the relative importance of genres within the schema must be linked to behavioral outcomes in question.

Following this approach, I showed that influential genres in the correlation network are also those for which there is a strong correlation between omnivorousness (in terms of actual choice behavior) and positive evaluation. I argued that this pattern of results can be accounted for if we read the schema correlation as a possible conduit for "spreading activation" of evaluations. This allows us to pick the genres the evaluation of which will be strongly coupled to the evaluation of other genres in the schema, from those whose evaluation will be loosely coupled. The within-schema correlation between omnivorousness at the level of consumption choices and the taste evaluation of each can then be used to define distinct omnivore/univore profiles for each class, thus providing schematic foundations for the claim that there are "varieties" of omnivorousness (and univorousness).

Despite these promising results, several limitations should be noted. Because the behavioral listening measures and the taste evaluations are self-reported concurrently in a cross-sectional design, we cannot rule out consistency bias—where respondents who express a liking for a genre are also more likely to report listening to it simply to maintain self-consistency. Consequently, these findings should be interpreted with care, and the "spreading activation" model should be understood as a useful methodological heuristic rather than a literal causal diagram of taste formation. Future longitudinal work would be necessary to establish a direct causal link between changes in schema organization and subsequent music consumption behavior.

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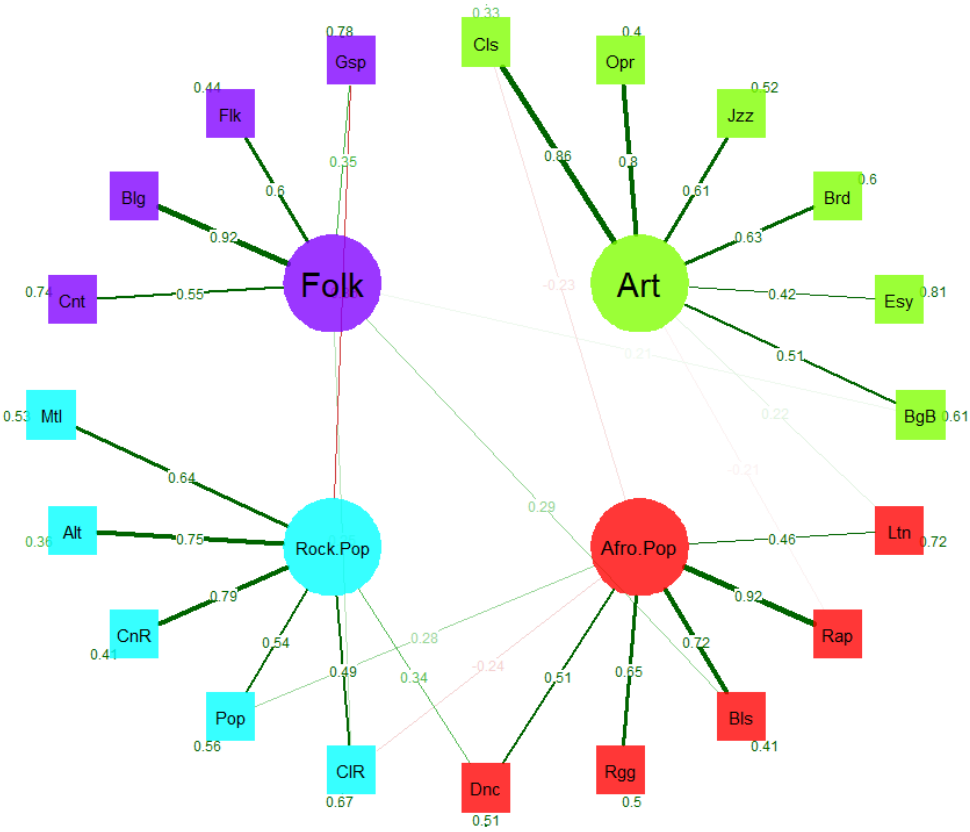
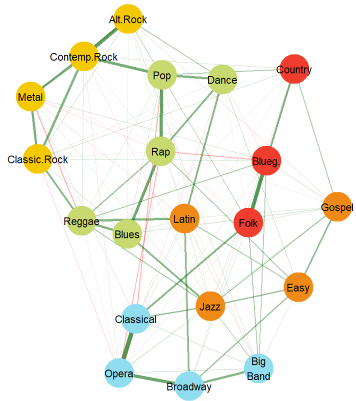
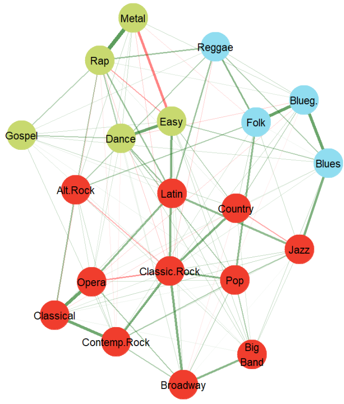


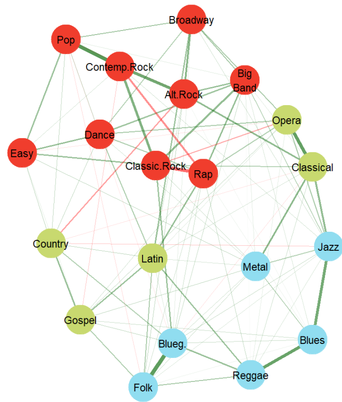
Figure 1: Behavioral Meta-Genre Typology Obtained from Factor Analysis of Musical Listening Choices. Circular Nodes are Latent Factors and Square Nodes are Genre Labels. Green Edges Indicating Positive Loadings on the Factor, Red Edges Indicate Negative Loadings. Transparency and thickness of edges is proportional to the magnitude of the loading. Legend: Cnt = Country, Blg = Bluegrass, Flk = Folk, Gsp = Gospel/Religious, Ltn = Latin/Spanish/Salsa, Rap = Hip Hop/Rap, Bls = Blues and R&B, Rgg = Reggae, Dnc = Dance/Electronic, Cls = Classical, Opr = Opera, Jzz Jazz, Brd = Broadway/Musicals, Esy = Easy Listening, Bgb = Big Band, ClR = Classic Rock/Oldies, Pop = Pop/Top 40, CnR = Contemporary Rock, Alt = Alternative Rock, Mtl = Hard Rock/Heavy Metal.



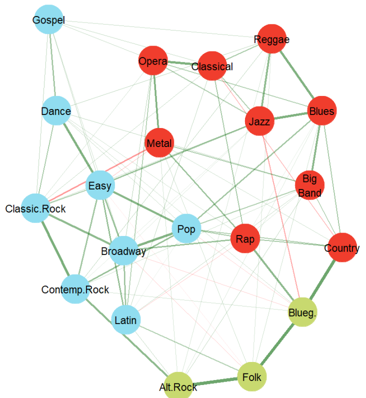
(a) Anything (but) Country or Rap
N = 481



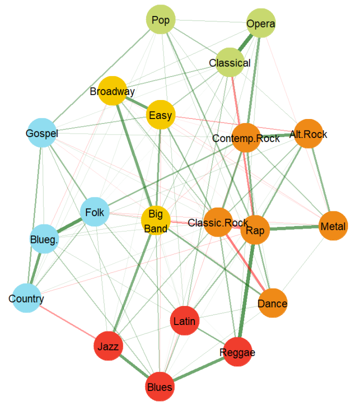
(b) Anything (but) Heavy Metal
N = 369



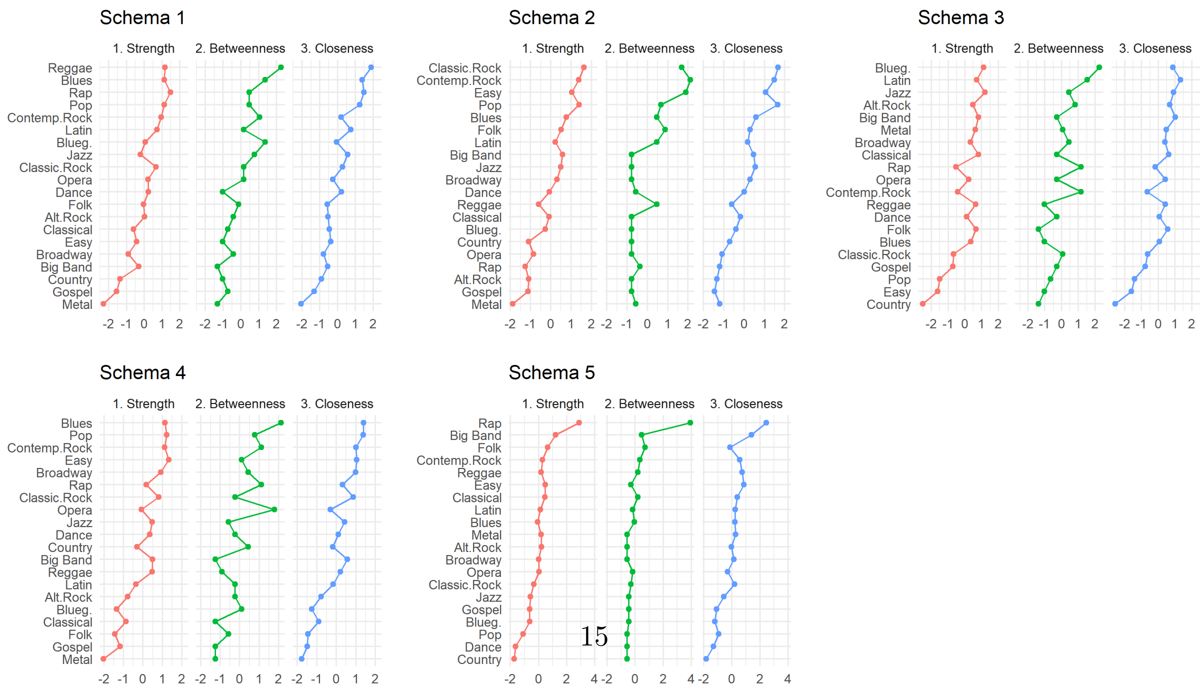
(c) Contemporary/Traditional
N = 574



(d) Omnivore/Univore
N = 419

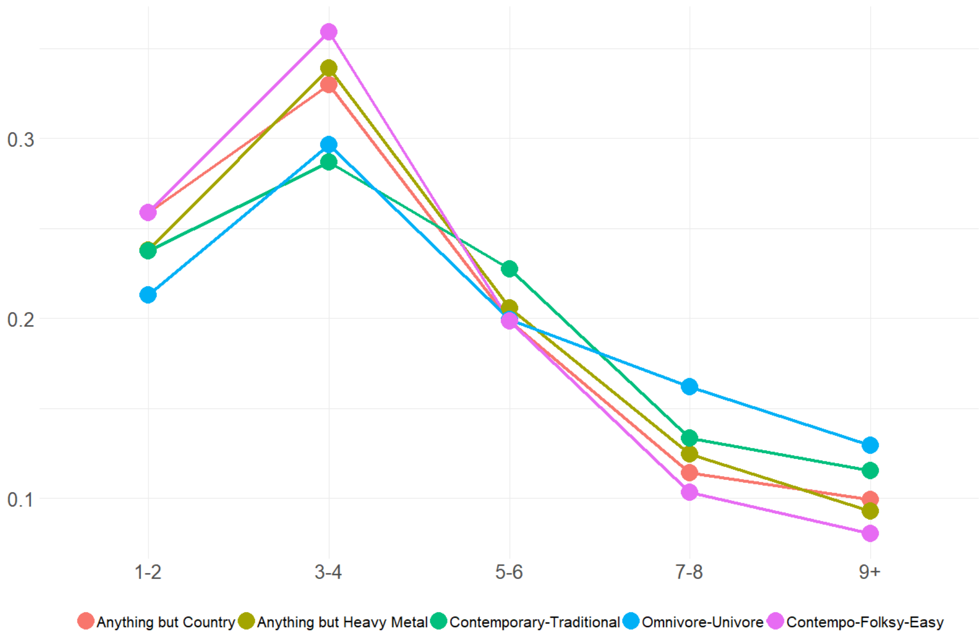


(e) Contempo/Folksy/Easy
N = 340

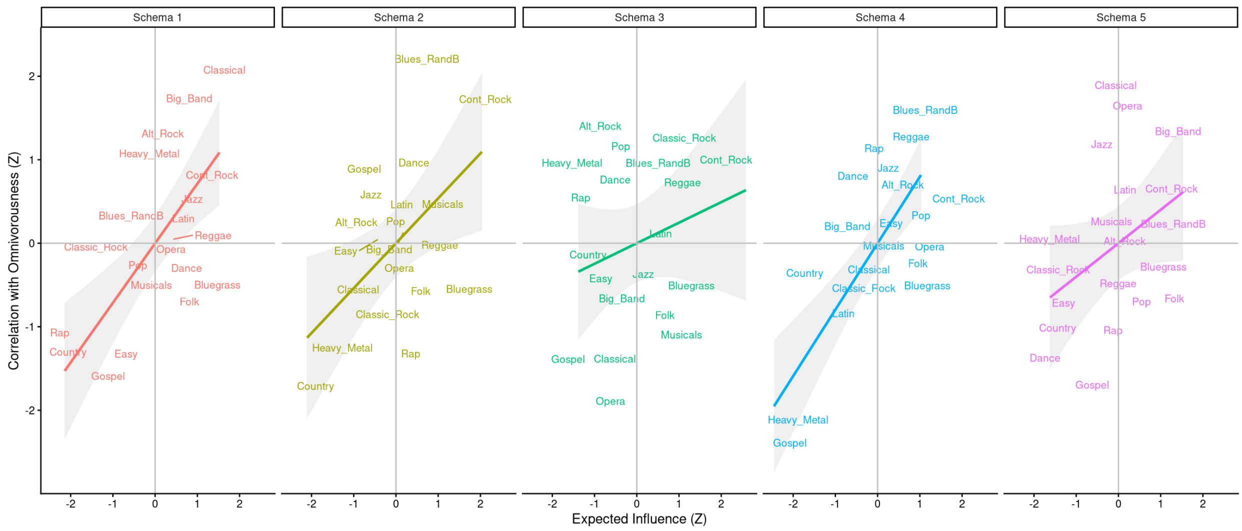


(f) Standardized Expected Influence Scores for Each Genre

Figure 2:
Top (a-e): Correlation Networks for each taste schema. Node color corresponds to the assignment of the genre to one of four latent factors based on independent consumption items (see Figure 1). Edge color indicates a positive (gray) or negative (red) correlation between genre evaluations within each class. Edge thickness is proportional to the absolute value of the polychoric correlation. Correlations that do not meet a statistical significance threshold ($p < 0.01$) are set to zero. Significant correlations



(a)



(b)

Figure 3:
Top: X-Axis is the number of genres respondents report listening to. Y-axis is the proportion of respondents in the sample who fall into that category.
Bottom: Y-axis is the (rescaled within taste schema) Spearman rank correlation between number of genres respondents listen to and their taste evaluation of that genre (higher values indicating the respondent likes the genre). X-axis is the (standardized at taste schema class mean) expected influence (EI_{gs}) of that genre in the correlation network for that schema (as given in figure 2f). Within each taste schema, the positive correlation between omnivorousness and evaluation increases as the centrality of that genre in the schema goes up. Spearman rank correlation coefficient per panel: Anything (but) Country or Rap: $r = 0.62, p < 0.01$; Anything (but) Heavy Metal: $r = 0.38$; Contemporary/Traditional: $r = 0.21$; Omnivore-Univore: $r = 0.68, p < 0.01$; Contempo/Folksy/Easy: $r = 0.40$.